Institution or Laboratory Name

Course or Research Group

Electrical Characterisation of a Resistor

Laboratory Report

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Abstract

This short example demonstrates a two-column scientific laboratory report. An elementary resistor is the illustrative system. All reported measurements and uncertainties are **synthetic**; no real experiment is described. Replace this summary with the objective, method and actual main finding of your laboratory work.

1 Introduction

Explain the scientific question and the physical principles used to interpret the experiment. For an ideal ohmic resistor, voltage V and current I are related by

$$V = IR, \tag{1}$$

where R denotes resistance. Equation (1) shows an ordinary numbered equation in a narrow column.

2 Experimental Methods

Describe the equipment, measurement sequence and uncertainties in sufficient detail that another reader can understand the procedure. In this fictional example, imagine a variable voltage source and a current meter connected to a nominal 1 kΩ resistor.

Assume standard uncertainties of 0.01 V and 0.03 mA for illustration. These values were not obtained by calibration. The distinction between precision and accuracy, as well as methods of uncertainty propagation, is discussed by Taylor [1997].

3 Results

Table 1 illustrates numerical formatting. The values are invented and should be replaced with genuine measurements before scientific use. In a compact report, tables should be limited to quantities that contribute directly to the argument.

Table 1: Synthetic observations; not experimental data.

V (V)	I (mA)
0.50	0.50
1.00	1.01
1.50	1.49
2.00	2.02
2.50	2.49
3.00	3.01

Figure 1 is generated directly in LaTeX and fits the width of one column. The line is an illustrative ideal relationship, not a fitted curve.

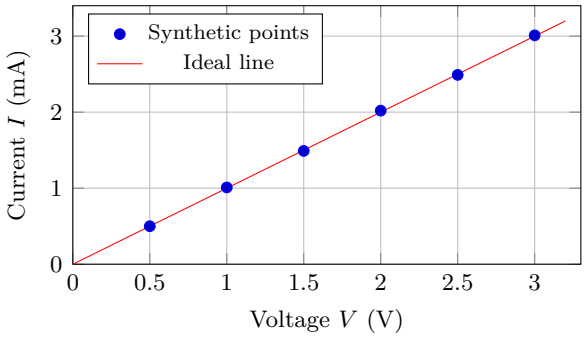


Figure 1: Illustrative current-voltage relationship using fictional points.

4 Discussion

Interpret the results and compare them with theoretical expectations. For example, a resistance can be calculated from a suitable current-voltage relationship. For a ratio $R = V/I$, a simple uncertainty expression with independent inputs is

$$\frac{u(R)}{R} \approx \sqrt{\left(\frac{u(V)}{V}\right)^2 + \left(\frac{u(I)}{I}\right)^2}. \tag{2}$$

Real data analysis also requires attention to correlated uncertainties, instrument offsets, calibration and the appropriateness of a linear model. In your own report, discuss limitations explicitly and avoid treating a visually straight line as proof that all assumptions are satisfied.

5 Conclusion

Summarise the scientific outcome, the principal uncertainty and one realistic improvement. Remove every fictional data point and statement before submitting a real laboratory report.

References

John R. Taylor. *An Introduction to Error Analysis: The Study of Uncertainties in Physical Measurements*. University Science Books, 2 edition, 1997.